# Choosing Table Primary Key

# Reminder: What is a primary key?

- Superkey:
  - A set of attributes such that …
  - If tuples have the same superkey values,
  - Then tuples are the same.
- Key: a minimal superkey
- Primary key: a chosen key

# Pragmatic goals for PK

- Unique
- Exists/known
- Immutable
- Simple/compact

# Two design questions

1. Natural vs. synthetic?
2. How to choose among multiple natural keys?

# Natural vs. synthetic?

**Natural**

- Meaningful data already included in DB needs

**Synthetic**

- Meaningless
- Made up for internal DB use

| Table | Primary key??? |
|---|---|
| Person | SSN, thumbprint |
| Course | Dept code + number |
| Team player | Jersey number |
| Campus building | Name, abbreviation |
| Library book | ISBN |

# Main tradeoffs

| | Natural | Synthetic |
|---|---|---|
| Already need data | Yes | No |
| Unique | Usually | Yes |
| Exists/known | Usually | Yes |
| Immutable | Usually | Almost always |
| Simple/compact | Usually | Yes |

# Natural vs. Synthetic: My recommendation

Use synthetic keys only as substitutes for complex natural keys.

# How to choose among natural keys?

Pick which best meets goals:

- Unique
- Exists/known
- Immutable
- Simple/compact

# How to generate synthetic keys?

- `SMALLSERIAL`, `SERIAL`, `BIGSERIAL` types
  - Auto-incrementing integers, fast
  - Insufficient for concurrent systems

- `UUID` type & `uuid_generate_v4()`
  - Unique even in concurrent systems